Linear equations

1 For each of the following statements:

i Express as an equation.

ii State whether the equation is linear.

a y is equal to 7 less than 2 groups of x.

b y equals x divided by 2 plus 8.

c y equals -1 divided by x plus 6.

d y plus the quotient of x and the square of -4 is equal to 14.

Tables of values

2 For each of the following tables representing a linear relationship:

i State what happens to the y-value when the x-value increases by 1.

ii Describe the rule between x and y in words.

iii Write the linear equation for the rule between x and y.

a

x	1	2	3	4	5
y	6	12	18	24	30

b

x	1	2	3	4	5
y	8	9	10	11	12

c

x	1	2	3	4	5
y	-7	-8	-9	-10	-11

d

x	1	2	3	4	5
y	-2	-6	-10	-14	-18

e

x	1	2	3	4	5
y	7	9	11	13	15

3 Consider the following table:

a For every 1 unit increase in the x-value, is there a consistent change in the y-value?

b State the change in y for every 1 unit increase in x.

x	9	18	27	36
y	-68	-131	-194	-257

c Explain why we can say that the relationship between x and y is linear.

d Write the linear equation that describes the relationship between x and y.

e Explain how the answer to part (b) helps in writing the rule for the linear relationship.

4 For each of the following tables, state whether the relationship between x and y is linear:

a

x	1	2	3	4	5
y	4	8	12	16	20

b

x	1	2	3	4	5
y	4	5	6	7	8

c

x	1	2	4	7	8
y	5	8	11	14	17

d

x	1	2	3	4	5
y	-1	3	7	11	15

e

x	1	2	3	4	5
y	5	1	-3	-7	-11

f

x	1	2	3	4	5
y	5	7	13	19	24

g

x	1	2	7	9	13
y	-3	-3	-3	-3	-3

h

x	5	10	15	20	25
y	15	30	45	60	75

i

x	-1	0	1	2
y	7	11	7	3

j

x	-1	0	1	2
y	3	7	11	15

k

x	-1	0	1	2
y	3	3	3	3

l

x	-1	0	1	2
y	3	7	11	13

m

x	3	3	3	3
y	0	3	6	9

n

x	0	3	6	9
y	-7	0	7	14

o

x	0	1	3	7
y	14	7	0	-7

p

x	-6	-5	-3	1
y	0	9	18	27

q

x	-6	-3	0	6
y	27	18	9	0

r

x	-6	-3	0	3
y	0	9	18	27

5 For each of the following tables:

i Use the pattern to find the value of y, when $x = 0$.

ii Write the linear equation that describes the relationship between x and y.

iii Hence, complete the table.

a

x	1	2	3	4	5	14
y	4	8	12	16	20	

b

x	1	2	3	4	5	30
y	-1	1	3	5	7	

c

x	-16	1	2	3	4	5
y		-2	-4	-6	-8	-10

d

x	-30	1	2	3	4	5
y		52	44	36	28	20

e

x	1	2	3	4	5	30
y	-13	-20	-27	-34	-41	

6 Given that the linear relationship between [redacted] y is in the form $y = mx + c$, for each of the following tables:

i Find the values of m and c.

ii Write the linear equation that describes the relationship between x and y.

iii Hence, complete the table.

a

x	0	1	2	3	4	5	21
y	0	2	4	6	8	10	

b

x	0	1	2	3	4	5	29
y	8	13	18	23	28	33	

c

x	0	1	2	3	4	5	21
y	-21	-16	-11	-6	-1	4	

d

x	0	1	2	3	4	5	65
y	24	21	18	15	12	9	

7 Find the linear equation between x and y for each of the following tables:

a

x	-1	0	1	2	3
y	5	2	-1	-4	-7

b

x	1	2	3	4	5
y	7	10	13	16	19

c

x	-8	-7	-6	-5	-4
y	-36	-31	-26	-21	-16

d

x	3	4	5	6	7
y	-12	-17	-22	-27	-32

Applications

8 The amount of medication M (in milligrams) in a patient's body gradually decreases over time t (in hours) according to the equation $M = 1050 - 15t$.

a After 61 hours, how many milligrams of medication are left in the body?

b How many hours will it take for the medication to be completely removed from the body?

9 A diver starts at the surface of the water and begins to descend below the surface at a constant rate. The table shows the depth of the diver over 5 minutes:

Number of minutes passed (x)	0	1	2	3	4
Depth of diver in metres (y)	0	1.4	2.8	4.2	5.6

a What is the increase in depth each minute?

b Write an equation for the relationship between the number of minutes passed (x) and the depth (y) of the diver.

c In the equation from part (b), what does the number in front of the x represent?

d Find the depth of the diver after 6 minutes.

e How long does the diver takes to reach 12.6 metres beneath the surface?

10 After Mae starts running, her heartbeat, in beats per minutes, increases at a constant rate.

a Write down the missing value from the table:

Number of minutes passed (x)	0	2	4	6	8	10	12
Heart rate (y)	49	55	61	67	73	79	

b What is Mae's resting heart rate?

c Find the change in y for every increase of one minute.

d Form an equation that describes the relationship between the number of minutes passed, x, and Mae's heartbeat, y.

e In the equation from part (d), what does the number in front of the x represent?

f Find Mae's heartbeat after twenty minutes.